B5. Condition Coverage and MC/DC

These two techniques examine the individual Boolean conditions inside decisions.

1. Condition Coverage

Condition Coverage checks whether every individual condition has been evaluated to both `True` and `False`.

Consider:

if decision == "WAITLISTED" and student.student_id not in course.waitlist:
course.waitlist.append(student.student_id)

There are two individual conditions:

- A: `decision == "WAITLISTED"`
- B: `student.student_id not in course.waitlist`

We need to test both conditions as True and False.

Test	A	B	Expected behavior
CC-01	True	True	Add student to waitlist
CC-02	True	False	Do not add duplicate student
CC-03	False	Not evaluated	Do not add student

In Python, `and` uses short-circuit evaluation. When A is False, B is not evaluated. So we need another test where A is True and B is False to exercise B's false outcome.

2. MC/DC (Modified Condition/Decision Coverage)

MC/DC checks whether each individual condition can independently change the final decision outcome.

Using the same expression:

A and B

We can use these two tests to prove that B independently affects the result:

Test	A	B	Final decision
MC-01	True	True	True
MC-02	True	False	False

Here, only B changes, and the final decision changes.

To prove that A independently affects the result:

Test	A	B	Final decision
MC-03	True	True	True
MC-04	False	True	False

Here, only A changes, and the final decision changes.

This is the basic idea of MC/DC: each condition must independently affect the final decision.